

VISHAY AGARWAL

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EDUCATION

Lynbrook High School

Relevant Coursework: AP Computer Science A, AP Computer Science Principles, Precalculus Honors

San Jose, CA
Expected Jun. 2029

EXPERIENCE

Berkeley Lab / Sonoma Water / USGS

Machine Learning Researcher | Scientific Software & Inference Systems

Berkeley, CA
2026 – Present

- Built production research infrastructure over 3.5M five-minute measurements from 35 river sites, with physics-constrained inference engines for phase lag, amplitude attenuation, depth structure, uncertainty, and source-linked evidence.
- Shipped reproducible analysis and validation pipelines generating 6,552 matched comparisons and figures used in 10+ technical reviews with engineers and research scientists across LBNL, Sonoma Water, and USGS.

AIEA Lab, University of California, Santa Cruz

Machine Learning Research Intern | Explainable AI & Autonomous Driving

Santa Cruz, CA
Jun. 2026 – Present

- Led a student research team building CARLA-based explainable ML for self-driving systems; reviewed PRs and implementations, integrated changes into a shared research codebase, and presented results to researchers.
- Built reproducible GPU training/evaluation infrastructure on the Nautilus Supercomputer, orchestrating compute through Kubernetes; ran 312,384 memory-strategy evaluations across 3 seeds with versioned logs, artifacts, and implementation commits.

Mathematical Foundations of Computer Science Lab, Chapman University

Research Intern | Spectral Graph Methods & Network Modeling

Orange, CA
Mar. 2026 – Present

- Built Python/PostgreSQL research infrastructure modeling 12.4M+ posts and 40M+ users with sparse graph representations, PageRank/eigenvector-style centrality, and network ranking methods; improved ingestion 30% and pipeline throughput 20%.

TECHNICAL SKILLS

Languages: Java, Ruby, JavaScript/TypeScript, Go, Python, C++, SQL, HTML/CSS

Frameworks: React, Next.js, FastAPI, Node.js, Electron, Three.js, PyTorch

Systems & Tools: Git/GitHub, code review, PR workflows, PostgreSQL, SQLite, MongoDB, Docker, Kubernetes, CUDA, Linux, REST APIs, WebSockets, Playwright, Vitest

Open Source: Contributions and PRs across Google DeepMind, OpenAI, Apple AI/ML Research, and Nous Research.

PROJECTS

Sift

Python, FastAPI, React, Three.js, SQLite, vector search

github.com/v1shay/sift
2026

- Built a spatial GitHub search engine over 10,223 repositories and 1M+ open items, combining vector retrieval, cosine similarity, graph clustering, and sub-5 ms search to surface contribution-ready codebases.

ArchLLM

C++20, CUDA, CMake, Kubernetes

github.com/v1shay/archLLM
2026

- Built a C++/CUDA inference system operating directly on GPU memory, reaching 92.4% on Microsoft STATE-Bench while reducing HBM pressure 30% and maintaining 95% memory-budget adherence.

EchoOS

Python, React, Electron, WebSockets, ElevenLabs

github.com/v1shay/echo-OS
2026

- Built a voice-controlled desktop interface enabling blind and mobility-impaired users to open apps, navigate files, browse the web, and execute computer actions entirely through speech.

ML-Labs

Python, React, scikit-learn, multi-agent orchestration

github.com/v1shay/ml-labs
2026

- Built ML infrastructure where 19 autonomous agents transform user data into trained models and production-ready APIs across 8 stages spanning ingestion, experimentation, validation, testing, and deployment.

Moonshot Robotics

TypeScript, React, Three.js, C++, MongoDB, reinforcement learning

github.com/v1shay/moonshot-robotics
2026

- Built an AI-native robotics engineering platform that autonomously sources hardware, designs components, assembles 3D systems, generates physics simulations, and trains robots with native C++ reinforcement learning.

idō

Python, Blender, OpenSCAD, agent tooling

github.com/v1shay/ido
2026

- Built an open-source 3D design harness enabling agents to generate parametric engineering assets across Blender and OpenSCAD, later powering Moonshot's physical-design layer; 3rd / 300+ at AWS × Anthropic.

AWARDS | \$7.73K+ EARNED ACROSS HACKATHONS

- 1st Place, ElevenLabs Voice Hackathon** — 1st / 120 teams (\$20K+ in prizes)
- 1st Place, ML Empowerment Build Challenge** — 1st / 600+ teams (\$50.4K prize pool)
- 3rd Place, AWS × Anthropic Harness Engineering Hackathon** — 3rd / 300+ participants (\$45K+ in prizes)
- 4th Place, YC × AWS Build YC's Next Unicorn Hackathon** — 4th / 1.1K+ teams
- 6th Place, Creator Colosseum Startup Competition** — 6th / 590+ teams
- 1st Place, International CCI Fair** — 1st / 1,400+ students worldwide